BASE CHANGE FOR THE UPPER SHRIEK ALONG A FLAT MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0E9U (sources/stacks-duality/pinned-original-source.tex line 4997).

Allowed standard prerequisites (editorial).

- The upper shriek through a compactification and the base change map for the right adjoints. Same chapter, Lemma labelled lemma-twisted-inverse-image and equation labelled equation-base-change-map, cited at proof lines 5015-5028. Define the map whose bijectivity is asserted.
- Affine locality of the right adjoint and the reduction to affine targets. Same chapter, Lemma labelled lemma-proper-noetherian and Remark labelled remark-check-over-affines, cited at proof lines 5036-5040. Reduce the isomorphism to affine Y and Y'.
- The base change statement in the affine flat case. Same chapter, Lemma labelled lemma-more-base-change, cited at proof lines 5040-5041. Settles the reduced case.

Source reference. Original reference 0F42: sources/stacks-duality/pinned-original-source.tex, source lines 3953–3953. Same-chapter pointer into the author environment situation at source line 3953; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original equation 0AA6: sources/stacks-duality/pinned-original-source.tex, source lines 1192–1192. Same-chapter pointer into the author environment equation at source line 1192; invoked at proof line 5024 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A9E: sources/stacks-duality/pinned-original-source.tex, source lines 653–653. Same-chapter pointer into the author environment

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lemma at source line 653; invoked at proof line 5026 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A9P: sources/stacks-duality/pinned-original-source.tex, source lines 1106–1106. Same-chapter pointer into the author environment lemma at source line 1106; invoked at proof line 5036 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original reference 0E9S: sources/stacks-duality/pinned-original-source.tex, source lines 1585–1585. Same-chapter pointer into the author environment remark at source line 1585; invoked at proof line 5038 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0AA8: sources/stacks-duality/pinned-original-source.tex, source lines 1636–1636. Same-chapter pointer into the author environment lemma at source line 1636; invoked at proof line 5040 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *In Situation 0F42 let*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{g} & Y \end{array}$$

be a cartesian diagram of FTS_S with g flat. Then there is an isomorphism $L(g')^* \circ f^! \to (f')^! \circ Lg^*$ on $D^+_{QCoh}(\mathcal{O}_Y)$.

Proof. Namely, because g is flat, for every choice of compactification $j : X \to \overline{X}$ of X over Y the scheme $\overline{X}$ is Tor independent of Y'. Denote $j' : X' \to \overline{X}'$ the base change of j and $\overline{g}' : \overline{X}' \to \overline{X}$ the projection. We define the base change map as the composition

$$L(g')^* \circ f^! = L(g')^* \circ j^* \circ a = (j')^* \circ L(\overline{g}')^* \circ a \longrightarrow (j')^* \circ a' \circ Lg^* = (f')^! \circ Lg^*$$

where the middle arrow is the base change map (0AA6) and a and a' are the right adjoints to pushforward of Lemma 0A9E for $\overline{X} \to Y$ and $\overline{X}' \to Y'$. This construction is independent of the choice of compactification (we will formulate a precise lemma and prove it, if we ever need this result).

To finish the proof it suffices to show that the base change map $L(g')^* \circ a \to a' \circ Lg^*$ is an isomorphism on $D^+_{QCoh}(\mathcal{O}_Y)$. By Lemma 0A9P formation of a and a' commutes with restriction to affine opens of Y and Y'. Thus by Remark 0E9S we may assume that Y and Y' are affine. Thus the result by Lemma 0AA8. □